

 Product Requirements Document (PRD)

Project Name: Prism IQ

Version: 1.0

Document Type: Product Requirements Document (PRD)

Prepared By: Prism IQ Development Team

Status: Draft v1.0

1. Executive Summary

Product Name

Prism IQ

Tagline

Transforming Data into Decisions.

Overview

Prism IQ is an intelligent end-to-end data analytics platform that enables users to import structured datasets from multiple sources, automatically clean and profile data, perform exploratory data analysis (EDA), build predictive machine learning models, generate business recommendations, and produce executive-ready reports—all within a unified web application.

The platform is designed to eliminate fragmented analytics workflows by combining data preparation, visualization, machine learning, and reporting into a single seamless experience.

Rather than requiring users to switch between spreadsheets, notebooks, visualization tools, and reporting software, Prism IQ provides a centralized analytics workspace suitable for students, analysts, researchers, and business professionals.

2. Problem Statement

Modern data analysis often requires multiple disconnected tools:

- Microsoft Excel
- Python scripts
- Jupyter Notebook

- Power BI
- Tableau
- Machine Learning notebooks
- Word or PowerPoint reports

This fragmented workflow creates several challenges:

- Manual data cleaning
- Repetitive preprocessing
- Slow insight generation
- Technical barriers for non-programmers
- Inconsistent reporting
- Poor collaboration
- Increased analysis time

There is a need for a unified platform capable of handling the complete analytics lifecycle from raw data to actionable business intelligence.

Prism IQ addresses this gap through intelligent automation while preserving transparency and user control.

3. Vision Statement

To build an intelligent analytics platform that transforms raw structured data into actionable business intelligence through automation, machine learning, and intuitive visualization.

4. Mission Statement

Enable users to make faster and more informed decisions by simplifying every stage of the data analytics workflow—from importing datasets to generating executive-ready business reports.

5. Product Objectives

Primary Objectives

- Support importing datasets from multiple sources
- Automatically profile uploaded datasets
- Detect common data quality issues
- Suggest and perform intelligent data cleaning

- Generate downloadable cleaned datasets
 - Perform Exploratory Data Analysis (EDA)
 - Allow user-selected prediction targets
 - Automatically detect ML problem type
 - Train multiple machine learning models
 - Select the best-performing model
 - Generate actionable business recommendations
 - Export professional PDF reports
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Secondary Objectives

- Modern and intuitive user interface
 - Beginner-friendly analytics workflow
 - Modular backend architecture
 - Reusable analytics pipeline
 - Scalable foundation for future releases
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6. Target Audience

Primary Users

Business Analysts

Need rapid insights and business reports.

Data Analysts

Need data cleaning, visualization, and predictive modeling.

Students

Need a practical platform to learn analytics and machine learning.

Researchers

Need efficient exploration of structured datasets.

Small and Medium Businesses

Need affordable business intelligence without enterprise software complexity.

7. Supported Dataset Categories (MVP)

Prism IQ Version 1 officially supports structured tabular datasets in the following categories:

- Sales & Retail
 - Customer Analytics
 - Marketing Campaigns
 - Finance & Banking
 - Stock Market
 - Weather Data
 - Education
 - Healthcare (non-sensitive structured datasets)
 - HR Analytics
 - Generic CSV/Excel datasets
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8. Supported Data Sources

Local File Upload

- CSV
 - Excel (.xlsx)
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URL Import

Extract structured HTML tables from webpages.

API Import

Support selected public APIs such as:

- Weather APIs
 - Financial/Stock APIs
 - Open Government Data APIs
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9. Product Scope (MVP)

The Minimum Viable Product includes:

Data Import

- CSV Upload
 - Excel Upload
 - API Import
 - HTML Table Extraction
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Data Profiling

Automatic generation of:

- Dataset overview
 - Missing values summary
 - Duplicate analysis
 - Data types
 - Column statistics
 - Memory usage
 - Cardinality analysis
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Data Cleaning

Automatically detect:

- Missing values
- Duplicate records
- Invalid data types
- Outliers (where applicable)
- Inconsistent categorical values

Users can preview cleaning actions before applying them.

A cleaned dataset can be downloaded after processing.

Exploratory Data Analysis (EDA)

Generate visual analytics including:

- Histograms
 - Bar Charts
 - Pie Charts
 - Line Charts
 - Scatter Plots
 - Box Plots
 - Heatmaps
 - Correlation Matrix
 - Distribution Analysis
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Predictive Analytics

The user selects the prediction target column.

The platform automatically:

- Detects Regression vs Classification

- Splits training/testing datasets
 - Trains multiple ML models
 - Evaluates model performance
 - Selects the best-performing model
 - Displays evaluation metrics
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Business Insights

Automatically generate:

- Trend analysis
 - Correlation highlights
 - Key observations
 - Business recommendations
 - Executive summary
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Dashboard

Interactive dashboard containing:

- Dataset Summary
 - Cleaning Results
 - EDA Visualizations
 - Model Performance
 - Business Insights
 - Prediction Results
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Report Generation

Generate:

- Executive PDF Report
 - Cleaned Dataset CSV
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10. Functional Requirements

The platform shall:

- Import supported datasets.
- Validate uploaded data.
- Generate automatic profiling reports.
- Suggest cleaning operations.

- Allow users to apply cleaning.
 - Generate cleaned datasets.
 - Produce EDA visualizations.
 - Support user-selected prediction targets.
 - Train multiple ML algorithms.
 - Compare model performance.
 - Generate business recommendations.
 - Export executive reports.
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11. Non-Functional Requirements

Performance

- Support datasets up to approximately **100,000 rows** for the MVP.
- Responsive user interface.
- Efficient backend processing.

Reliability

- Graceful error handling.
- Data validation before processing.
- Safe handling of failed imports.

Scalability

Architecture should support future expansion to:

- Authentication
- Multi-user projects
- Cloud storage
- Scheduled pipelines
- Additional ML algorithms

Usability

- Beginner-friendly interface
- Minimal learning curve
- Guided workflow

Maintainability

- Modular architecture
 - Clean codebase
 - Comprehensive documentation
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12. Success Metrics

The MVP is considered successful if users can:

- Import supported datasets.
 - Successfully clean datasets.
 - Download cleaned data.
 - Generate meaningful visualizations.
 - Train predictive models.
 - Receive actionable business recommendations.
 - Export executive reports.
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13. Out of Scope (Version 1)

To meet the project timeline, the following features are excluded:

- User Authentication
- Multi-user collaboration
- Real-time streaming analytics
- AutoML frameworks
- LLM-powered recommendations
- Time-series forecasting
- Database import connectors
- Role-based permissions
- Scheduled jobs
- Mobile application

These are planned for future releases.

14. Technology Stack (Approved)

Frontend

- React
- Vite
- TypeScript
- Tailwind CSS
- shadcn/ui
- Recharts

- Plotly
 - React Router
 - TanStack Query
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Backend

- FastAPI
 - SQLAlchemy
 - Pandas
 - NumPy
 - Scikit-learn
 - ReportLab
 - BeautifulSoup4
 - HTTPX
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Database

PostgreSQL

Reason:

- Industry-standard relational database
 - Strong SQLAlchemy support
 - Better scalability than SQLite
 - Suitable for future authentication, project history, and multi-user expansion
 - Excellent portfolio value
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15. Risks & Mitigation

Risk	Mitigation
Large datasets increase processing time	Use efficient Pandas operations, sampling for

	visualizations, and background processing where needed
API failures	Validate responses and provide graceful fallback messages
Unsupported webpage structures	Restrict URL import to standard HTML tables for MVP
ML model underperformance	Train multiple models and automatically select the best-performing one

Tight deadline	Prioritize MVP features and defer advanced capabilities to future versions
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16. Future Roadmap

Potential future enhancements include:

- User authentication & authorization
- Saved workspaces and project history
- Database connectors (MySQL, PostgreSQL, SQL Server)
- Explainable AI (XAI)
- Automated pipeline scheduling
- Natural language querying
- Time-series forecasting
- Collaboration and sharing
- Cloud deployment with monitoring
- Plugin architecture for additional data sources

17. Acceptance Criteria

The MVP will be accepted when:

- All supported import methods function correctly.
- Automatic profiling is generated.
- Cleaning suggestions are available and applicable.
- Cleaned datasets can be downloaded.
- EDA visualizations are generated successfully.
- Users can train predictive models using a selected target.
- The best-performing model is identified automatically.

- Business insights and recommendations are generated.
 - Executive PDF reports are downloadable.
 - The application demonstrates stable performance on representative datasets.
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